3.2 : Multiple linear regression

Why? Simple linear regression so far can accommodate multiple predictors

A motivating example using the data for the Green Member example from 3.1 linear regression.

Example includes recap from previous one-variable model including the coefficient estimates and p-value labels from CO2 ~ Income.

Also the graph for the linear model.

Set up: Training data $\{(x_1, y_1) \dots (x_n, y_n)\}$, this time

$x_i = (x_{i1}, \dots, x_{ip})$ is a vector with p predictors

y_i = response variable associated to x_i.

Example: model education with example variables and variation

Multiple linear regression model has the form

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon,$$

X_j = j-th predictor

β_j = parameter, quantifies association between variable and response.

example:

y = CO2-reduction (response variable)

$p = 3$ (number of predictors)

X_1 = process

X_2 = efficiency } predictors

X_3 = offset

CO2-reduction $= \beta_0 + \beta_1$ process $+ \beta_2$ efficiency $+ \beta_3$ offsets

Question: for a point (b_1, b_2, b_3) with data, how would you calculate the CO2-reduction

Estimating the regression coefficients

The coefficients $\beta_0, \dots, \beta_p$ are unknown and we need to estimate them. Parameter all the estimates under note: $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$. With these estimates we can use the formula

$$y = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p = \hat{f}(x_1, \dots, x_p)$$

to make prediction or inference questions.

Parameters are estimated in the same least squares approach:

we find $\hat{\beta}_0, \dots, \hat{\beta}_p$ that minimizing the residual squared sum RSS:

$$RSS = \sum_{i=1}^{n} (y_i - \hat{f}(x_i))^2$$

$$= \sum_{i=1}^{n} (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip}))^2$$

$$= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})^2$$

In this formula the $\hat{\beta}_j$ are unknown variables, we find the coefficient estimates $\hat{\beta}_j$ by minimizing it.

A cool abstract example would be to show how this looks like in two variables visually, and show how is a combination of parameters that minimizes this.

* This would also be good to explain in the parameter finding for the one variable.

How to interpret $\hat{\beta}_j$: the average effect on y if a one unit increase in x_j, holding all other predictors fixed.

Go back to our CO2-example and have a

- Ideally, we would get results similar to table 3.4

Exercise: how would you interpret the coefficients?

Comparing multivariate regression with individual regressions.

Now we need to continue the example now starting the coefficients from running individual regressions on each variable compared to multivariate regression.

Main points to examplify:

- Simple and multiple regression coefficients can be quite different.

Why?

- simple regression: coefficients ignore the existence of any other variables
- multiple regression: coefficients represents average change when all the other variables are fixed.

Pay a walk-out that one would suggests no relationship, while the other one does?

look at correlation matrix, table 3.5.

We also need to compare table 3.4 (coefficient, std error, t-statistic, p-value) for the multivariate regression and explain that the p-value here represents whether each predictor is associated to the response, when other predictors are held fixed.

↳ partial effect of adding that variable to the model."

Students might ask: should we run a correlation matrix before?

Example: share attacks, ice cream sales, temperature.

Question we can gain insight into

Is there a relationship between response and predictors?

Linear regression recap.

In the simple linear regression: $Y \approx \beta_0 + \beta_1 X_1$, so no relationship happens when $\beta_1 = 0$.

$H_0: \beta_1 = 0$

$H_a: \beta_1 \neq 0$

We test how far $\hat{\beta}_1$ is from 0 relative to its uncertainty: t-statistic

we ask what is the probability to observe this t value or larger if $\beta_1 = 0$, we get a p-value. Small p-value ⇒ there is evidence of a relationship.

In multiple regression: $Y \approx \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$, so no relationship happens when $\beta_1 = \dots = \beta_n = 0$ (all coefficients are 0).

So we use the null hypothesis

$$H_0: \beta_1 = \beta_2 = \dots = \beta_n = 0$$

with the alternative

H_a: at least one β_j is not 0

} I could make this into a check-in, or at least asking what would be H_0

We compute the F-statistic

$$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)}$$

Practical questions: intuitively, what is this measuring, just as the t-statistic is measuring std deviations from 0 ≠ this something similar?

Interpreting the F-statistic

- close to 1: expected when there is no relationship between Y and X_j. "at least one of the predictors X_j must be related to the response Y"
- greater than 1: expected when H_a is true
  - when n is large, an F-statistic slightly larger than 1 might provide evidence against H_0
  - when n is small, we need a large F-statistic to reject H_0

Potential question: could the F-statistic be negative?

how big is big for n? how small is small?

We can calculate a p-value associated with the F-statistic, naturally

"what is the probability of having this F statistic or ... [make our language matching the t-test p-value explanation]"

A small p-value: reject H_0.

Big p-value: cannot reject H_0.

Check in with example data and p-value.

Why do we need an F-statistic and the p-value if we already have p-values for each predictor?

[Here's a nice discussion about interpreting p-values vs. F-statistic in the context of a large number of predictors, but I will skip it (page 89) this could be re-discussed in larger discussions]

Deciding on important variables

So when we perform a multiple regression:

1) Fit the model
2) Compute F-statistic and associated p-value
3) Check the individual p-values for predictors.

If there's evidence that some predictors are associated with the variables, then we need to ask: which?

Remember: for large number of predictors p, individual p-values might have false positives.

Question from students: how large would p be so we worry about this?

Determining which predictors are associated with the response to fit a single model using only those predictors is called variable selection.

Q3: same as feature selection?

Trying out all possible combination of predictors is computationally unfeasible.

- Forward selection

Remember

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \text{residual sum of squares}$$

* I would have called this sum of squared residuals.

1. Start with a model that has an intercept but no predictors
   Q3: how does this look like, how do we implement it?
2. Fit simple linear regressions with a single variable for each predictor. Out of the p models add the variables that has lowest RSS.
3. Run a 2-variable model with the first selected variable and all the other variables. The second variable is the one with lowest RSS.
   * I want to have these variables selection methods as a lab or homework.

- Backward selection
- Mixed selection

Model fit

We still use the residual squared error RSE = formula and the R^2 = coefficient of determination = formula.

R^2 close to 1 ⇒ model explains a large portion of ... (use same language as linear regression)

Discussion with example about how R^2 will always increase with number of variables and we may want to be aware of that.

Predictions

Qualitative predictors.

Non-linear relationships

* Assign potential problems as reading and some kind of exercise. Have 6 sections.